higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRONICS
NQF LEVEL 2

18 NOVEMBER 2019

This marking guideline consists of 8 pages.

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QUESTION 1

1.1 D
1.2 B
1.3 C
1.4 B
1.5 C
1.6 A
1.7 B
1.8 B
1.9 B
1.10 B

(10 × 1) **[10]**

QUESTION 2

2.1
- Permanent magnet – its magnetic properties exist when the magnet is magnetised.

OR

- Magnet that stays magnetic all the time.
- Electromagnet – Its magnetic properties exist when electric current is applied to it. (2)

2.2
- Forefinger points in the direction of the field
- Second finger points in the direction of the current
- Thumb points in the direction of the force (3)

2.3
- The bigger the area, the lesser the resistance.
- Increasing the temperature, increases the resistance of the material. (2)

2.4

$$\frac{1}{R_P} = \frac{1}{10} + \frac{1}{20}$$

$$= \frac{20+10}{200} \text{ or } R_P = \frac{10\times 20}{10+30} \quad ✓$$

$$= \frac{200}{30}$$

$$R_P = 6.667\,\Omega \quad ✓$$

$$R_T = 25 + 6.667$$

$$= 31.667\,\Omega \quad ✓$$

(3)
[10]

QUESTION 3

3.1 Forward bias refers to when the P-type side of the diode is connected to a higher potential than the N-type side.

OR

Forward bias refers to when applied voltage causes the current to flow from the N-side to the P-side of the diode.

OR

Forward bias refers to the process where barrier resistance is decreased to allow current to flow.

OR

Forward bias refers to when a relatively small voltage of approximately 0,3 V (germanium material) or 0,7 V (silicon material) is applied across a diode in such a way that it allows current to flow from its anode to its cathode. (Any 1 × 1) (1)

3.2 Free electrons can move freely from one atom to another. (1)

3.3
- Brown
- Red
- Red
- Silver (4)

3.4 When a small amount of trivalent impurity✓ is added to a pure germanium or silicon atom crystal to produce a large number of holes.✓ (Any 2 × 1) (2)

3.5 An increase in temperature increases the number of carriers✓ in a semiconductor material, which leads to an increase in conductivity.✓

OR

An increase in temperature in a semiconductor material decreases the resistance,✓ which, in turn, allows more current to flow.✓ (2)

[10]

QUESTION 4

4.1
- Live circuit – current is flowing in the circuit.
- Dead circuit – no current is flowing in the circuit. (2)

4.2 To protect the consumer from an electric shock✓ and to avoid damages to electrical equipment.✓

OR

To provide an alternative path for dangerous currents to flow so that accidents due to electric shock✓ and damage to equipment can be avoided.✓ (2)

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4.3
- Children chewing on cords or sticking objects into outlets.
- Poorly functioning appliances or equipment.
- Frayed cords, exposed wires, missing prongs or plugging many cords into one outlet.
- Power lines that are down near water or machinery.
- Lightning. (Any 2 × 1) (2)

4.4
- Any electric shock if a woman is more than 20 weeks' pregnant.
- Burns with increasing redness, soreness and drainage.
- Any noticeable burn to the skin.
- Any period of unconsciousness.
- Any numbness, tingling, paralysis, vision, hearing or speech problems.

(Any 4 × 1) (4)
[10]

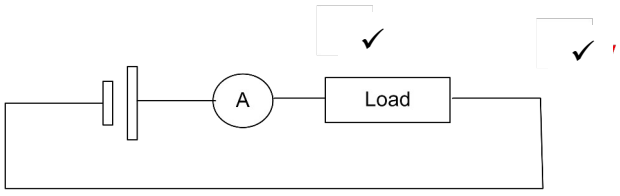
QUESTION 5

5.1 Voltage is directly proportional to current (an increase in voltage leads to an increase in current)✓ and resistance is inversely proportional to current (the higher the resistance, the lower the current).✓ (2)

5.2
$$I = \frac{V}{R}$$
$$= \frac{15}{10} \quad ✓$$
$$= 1.5\,A \quad ✓$$

(2)

5.3



(2)

5.4
- An electric shock
- Injury
- Death
- Damage to property
- Loss of money
- Loss of time
- Inaccurate readings from the instrument (Any 4 × 1) (4)

[10]

QUESTION 6

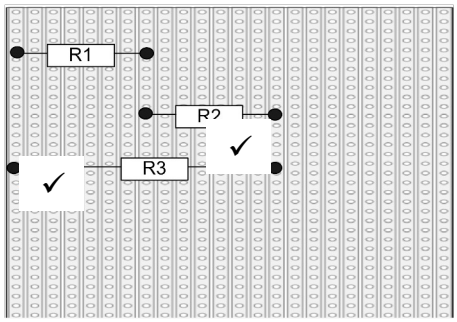
6.1 False
6.2 False
6.3 True
6.4 True
6.5 False
6.6 False
6.7 False
6.8 False
6.9 True
6.10 True

(10 × 1) **[10]**

QUESTION 7

7.1 A breadboard does not require soldering✓ and it can be used to build temporary circuits.✓ (2)

7.2



(3)

7.3
- An ammeter should not be connected across R1, but in series with R1.
- The transistor is supposed to be NPN and not PNP.
- The capacitor connected in the circuit is polarised instead of non-polarised.
- The value of the capacitor is in nano-Faraday instead of micro-Faraday. (4)

7.4 Oscilloscope (1)
[10]

QUESTION 8

8.1 $= 1\times 2^3 + 0\times 2^2 + 0\times 2^1 + 1\times 2^0$

$= 8 + 0 + 0 + 1$ ✓

$= 9_{10}$ ✓

8.2

Carry
1 1 1 1
1 0 0 1
+ 1 1 0 0 1
1 0 0 0 1 0

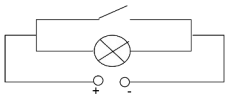
✓ ✓

8.3

1 1 0 0 1
- 1 0 0 1
1 0 0 0

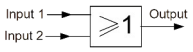
✓ ✓

8.4 ✓



✓

8.5 ✓



OR gate ✓

(5 × 2) **[10]**

QUESTION 9

9.1 Discrete inputs:

- Push buttons
- Limit switches
- Float switches
- Photoelectric switches
- Proximity switches (Any 1 × 1)

Analogue inputs:

- Sensors (temperature, pressure, speed, etc.)
- Weight scales
- Level transmitter (Any 1 × 1) (2)

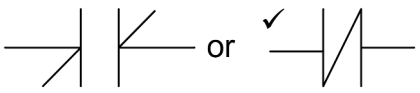
9.2 Use specific software to create and make changes to a program,✓ then use the connector cable to connect the computer to the PLC unit to transfer data.✓ (2)

9.3

- Calculate how much power the PLC, including the peripherals, need by referring to a manufacture's manual.
- Look up how much power the PLC uses and how much power each input and output that you attach uses.
- Add up all the values to determine the total power required.
- Use the correct transformer to produce the correct amount of power.
- A protective device and an insulating device must be provided.

(Any 4 × 1) (4)

9.4 Normal closed input De-energised output





(2)
[10]

QUESTION 10

10.1 Wavelength is the distance between two consecutive points on a wave, which are in phase.

OR

Wavelength is the distance between similar points on two back-to-back waves.

OR

Wavelength is the distance between two successive crests.

10.2

- Amplitude modulation (AM) involves the modulation of the carrier signal according to the amplitude of the baseband signal.
- Frequency modulation (FM) involves the modulation of the carrier signal according to the frequency of the baseband or information signal.

10.3 Signal-to-noise (SNR) ratio, which is the ratio of signal power to noise power, is often expressed in decibels.

OR

SNR is the measure of signal strength relative to the background noise.

10.4 PSTN – Public Switched Telephone Network
PABX – Private Automatic Branch Exchange

10.5 10.5.1 WAN
10.5.2 LAN

(5 × 2) **[10]**

TOTAL: 100

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